- **Internship Name : Data Analysis**
- **Task Name: Task 9: Handling Missing Values and Removing Duplicates using Pandas**
- **Your Name : Shubhangi Naresh Khonde**
- **Internship Organization :**
- **Date:**

1. Objective

To identify missing values (NaN) and duplicate records in a messy dataset, and clean them using the Pandas functions isnull(), fillna(), dropna() and drop_duplicates().

2. Tools Used

- Python 3.13
- Pandas and NumPy libraries
- Visual Studio Code (Jupyter Notebook)

3. Dataset Description

The sample dataset contained 8 rows and 4 columns: **Name, Age, City and Salary**.

It had the following data quality problems:

- **Missing values:** Name (1), Age (2), City (1), Salary (1), 5 in total
- **Duplicate records:** the entries for “Sneha”, “Rahul” and “Amit” were repeated. One “Amit” entry also had an extra trailing space ("amit "), which made it look different from the original.

```
##Step 1: Create Messy sample dataset
```

```
import pandas as pd
import numpy as np

data = {
    "Name":   ["Amit", "Sneha", "Rahul", "Sneha", "Priya", "amit ", None, "Rahul"],
    "Age":    [25, 28, np.nan, 28, 22, 25, 30, np.nan],
    "City":   ["Pune", "Mumbai", "Nagpur", "Mumbai", None, "Pune", "Nashik", "Nagpur"],
    "Salary": [50000, 62000, 45000, 62000, np.nan, 50000, 58000, 45000],
}
df = pd.DataFrame(data)
print(df)
```

✓ 3.6s

```
    Name   Age    City   Salary
0   Amit  25.0    Pune  50000.0
1  Sneha  28.0  Mumbai  62000.0
2  Rahul   NaN  Nagpur  45000.0
3  Sneha  28.0  Mumbai  62000.0
4  Priya  22.0     NaN      NaN
5  amit   25.0    Pune  50000.0
6    NaN  30.0  Nashik  58000.0
7  Rahul   NaN  Nagpur  45000.0
```

4. Methodology

Step 1: Detecting Missing Values

df.isnull() and df.isnull().sum() were used to find how many values were missing in each column.

```
##Step 2: Missing values shodha (isnull)
```

```
print(df.isnull())
print(df.isnull().sum())
print(df.info())
```

✓ 0.0s

```
    Name    Age   City  Salary
0  False  False  False   False
1  False  False  False   False
2  False   True  False   False
3  False  False  False   False
4  False  False   True    True
5  False  False  False   False
6   True  False  False   False
7  False   True  False   False
Name      1
Age       2
City      1
Salary    1
dtype: int64
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    7 non-null      str
 1   Age     6 non-null      float64
 2   City    7 non-null      str
 3   Salary  7 non-null      float64
dtypes: float64(2), str(2)
memory usage: 388.0 bytes
```

Step 2: Handling Missing Values

Column	Method Used	Reason
Age	fillna(median)	The median is less affected by outliers
Salary	fillna(mean)	The column is numeric, so the average is a suitable replacement
City	fillna("Unknown")	The column is text, so mean or median does not apply
Name	dropna(subset=["Name"])	Name is the key identifier, and a record without it has no meaning

Step 3: Cleaning Text Data

str.strip().str.title() was applied to the Name column to remove extra spaces and standardize capitalization. Because of this, "amit " and "Amit" were correctly recognized as the same entry.

Step 4: Removing Duplicates

- df.duplicated().sum() counted the duplicate rows (**3** were found).
- df.drop_duplicates() removed the duplicate rows.

```
df["Name"] = df["Name"].str.strip().str.title()

print(df.duplicated().sum())
print(df[df.duplicated(keep=False)])

df = df.drop_duplicates()
df = df.drop_duplicates(subset=["Name"],keep="first")
```

✓ 0.0s

```
3
    Name   Age    City   Salary
0   Amit  25.0    Pune  50000.0
1  Sneha  28.0  Mumbai  62000.0
2  Rahul  26.5  Nagpur  45000.0
3  Sneha  28.0  Mumbai  62000.0
5   Amit  25.0    Pune  50000.0
7  Rahul  26.5  Nagpur  45000.0
```

5. Results

Check	Before Cleaning	After Cleaning
Number of rows	8	4
Total missing values	5	0
Duplicate rows	3	0

```
##Final check
```

```
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.shape)
```

[7] ✓ 0.0s

```
Name      1
Age       0
City      0
Salary    0
dtype: int64
0
(5, 4)
```

6. Observations

- Without text cleaning, duplicates would have been missed because "Amit" and "amit " look like different values to Pandas.
- fillna() preserved the data by replacing missing values, while dropna() removed records that were unusable because a key column was empty.
- The order of operations matters: missing values and text formatting should be fixed before removing duplicates.

7. Conclusion

The messy dataset was cleaned using isnull(), fillna(), dropna() and drop_duplicates(). The final dataset has no missing values and no duplicate rows, so it is ready for analysis. Data cleaning is a critical first step in any data analysis workflow, because inaccurate data leads to inaccurate results.